

Your Thesis Title Here

Your Name

A thesis submitted in partial fulfillment
of the requirements for the degree of
Master of Science

Department Name

University Name

Month Year

Abstract

Your Thesis Title Here

Your Name

Write your abstract here. The abstract should provide a concise summary of your thesis, including the research problem, methodology, key findings, and conclusions. Typically, an abstract is between 150-300 words and should be written in a clear, accessible manner that allows readers to quickly understand the essence of your work.

This template uses double spacing for the abstract text. You can modify the spacing if required by your institution's guidelines.

Remember to replace all placeholder text with your actual content.

Abstract (Italian)

Il Titolo della Tua Tesi

Your Name

Scrivi qui il tuo abstract in italiano. L'abstract dovrebbe fornire un riassunto conciso della tua tesi, includendo il problema di ricerca, la metodologia, i risultati principali e le conclusioni.

Se non hai bisogno di un abstract in italiano, puoi rimuovere questo file e commentare la riga corrispondente nel file Thesis.tex principale.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
ML	Machine Learning

LIST OF SYMBOLS

- λ Lambda indicates usually an eigenvalue in linear algebra
- π A mathematical constant whose value is the ratio of any circle's circumference to its diameter

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Chapter 1

Introduction

This is the introduction chapter of your thesis. Here you should:

- Introduce the research topic and its context
- Explain the motivation for your research
- State the research problem or question
- Outline the objectives of your study
- Describe the structure of the thesis

1.1 Background

Provide background information that helps readers understand your research topic. This section should establish the context and importance of your work.

You can use **bold text**, *italic text*, and `monospace text` for emphasis.

1.2 Research Problem

Clearly state the research problem or question that your thesis addresses. Explain why this problem is important and worth investigating.

1.3 Research Objectives

List the specific objectives of your research. For example:

1. Objective 1: To investigate...

1.4. Thesis Structure

2. Objective 2: To develop...
3. Objective 3: To evaluate...

1.4 Thesis Structure

Provide an overview of how the thesis is organized. For example:

Chapter 2 presents the literature review and theoretical framework. Chapter 3 describes the methodology used in this research. Chapter 4 presents the results and analysis. Finally, Chapter 5 concludes the thesis with a discussion of findings and future work.

1.5 Example of Citations

This template is set to the manual bibliography (`useBibTeX` is `false` in `config.tex`), so references are written out by hand in `Bibliography.tex` and cited in the text by author and year, for example “as shown by Author (2023)”.

Set `useBibTeX` to `true` to switch to `References.bib`. The `\cite` command then becomes available and this paragraph is replaced by a worked example, without any other change to the document.

1.6 Example of Abbreviations and Symbols

Entries defined in `Lists.tex` only show up in the List of Abbreviations and the List of Symbols once they are actually used in the text. The first use of an acronym is expanded automatically, Artificial Intelligence (AI), and every later use is short: AI. The same applies to Machine Learning (ML) and Application Programming Interface (API).

Symbols behave the same way: π and λ .

1.7 Example of Figures

You can include figures in your thesis. Here’s an example:

Figure 1.1: Example figure caption. Always provide descriptive captions for your figures.

You can reference figures using `ref` command: see Figure 1.1.

For side-by-side panels use the `subfigure` environment from `subcaption`. Each panel needs an explicit width and carries its own `caption`; `ref` then yields ‘1.1a’, ‘1.1b’ and so on.

1.8 Example of Tables

Tables can be included as follows:

Table 1.1: Example table showing sample data

Column 1	Column 2	Column 3
Row 1	Data	123
Row 2	More data	456
Row 3	Even more	789

Reference tables like this: see Table 1.1.

1.9 Example of Equations

Mathematical equations can be included inline like $E = mc^2$ or as display equations:

$$\int_a^b f(x) dx = F(b) - F(a) \tag{1.1}$$

Reference equations using: see Equation 1.1.

Chapter 2

Literature Review

This chapter presents a comprehensive review of the existing literature related to your research topic. The literature review should:

- Identify key theories, concepts, and frameworks
- Summarize previous research findings
- Identify gaps in the current knowledge
- Justify the need for your research

2.1 Theoretical Framework

Describe the theoretical framework that underpins your research. Explain the key theories and concepts that guide your study.

2.1.1 Key Concepts

Define and explain the main concepts used in your research. For example:

Concept 1: Definition and explanation of the first key concept.

Concept 2: Definition and explanation of the second key concept.

2.1.2 Theoretical Models

Discuss relevant theoretical models that inform your research approach.

2.2 Previous Research

Review and synthesize previous research in your field. Organize this section thematically or chronologically, depending on what makes most sense for your topic.

2.2.1 Research Area 1

Discuss research related to the first major area of your study. Cite relevant sources and explain their findings.

2.2.2 Research Area 2

Discuss research related to the second major area of your study.

2.3 Research Gap

Based on your literature review, identify the gaps in current knowledge that your research aims to address. Explain how your work will contribute to filling these gaps.

2.4 Summary

Summarize the key points from your literature review and explain how they lead to your research questions or hypotheses.

Chapter 3

Methodology

This chapter describes the research methodology used in your study. It should provide enough detail for others to understand and potentially replicate your work.

3.1 Research Design

Describe the overall research design and approach. Explain whether your research is:

- Qualitative, quantitative, or mixed methods
- Experimental, observational, or theoretical
- Exploratory, descriptive, or explanatory

Justify why this approach is appropriate for your research questions.

3.2 Data Collection

Describe how you collected your data. Include information about:

3.2.1 Data Sources

Explain where your data comes from. This could include:

- Primary data (surveys, experiments, interviews)
- Secondary data (existing datasets, literature)
- Simulations or computational models

3.2.2 Sampling Strategy

If applicable, describe your sampling strategy:

- Population and sample size
- Sampling method (random, stratified, convenience, etc.)
- Inclusion and exclusion criteria

3.2.3 Data Collection Instruments

Describe the tools and instruments used for data collection (e.g., questionnaires, measurement devices, software).

3.3 Data Analysis

Explain the methods used to analyze your data:

3.3.1 Statistical Methods

If applicable, describe the statistical methods used:

- Descriptive statistics
- Inferential statistics
- Hypothesis testing procedures
- Software tools used (R, Python, SPSS, etc.)

3.3.2 Qualitative Analysis

If applicable, describe qualitative analysis methods:

- Coding procedures
- Thematic analysis
- Content analysis

3.4. Validity and Reliability

3.4 Validity and Reliability

Discuss measures taken to ensure the validity and reliability of your research:

- Internal validity
- External validity
- Reliability of measurements
- Potential sources of bias and how they were addressed

3.5 Ethical Considerations

Describe any ethical considerations relevant to your research:

- Informed consent procedures
- Data privacy and confidentiality
- Ethical approval (if required)

3.6 Limitations

Acknowledge any limitations of your methodology and explain how they might affect your results.

Chapter 4

Conclusion

This final chapter summarizes your research, discusses its implications, and suggests directions for future work.

4.1 Summary of Findings

Provide a concise summary of your main findings. Revisit your research questions or objectives and explain how your results address them.

Key findings include:

- Finding 1: Brief description
- Finding 2: Brief description
- Finding 3: Brief description

4.2 Discussion

Discuss the implications of your findings:

4.2.1 Theoretical Implications

Explain how your findings contribute to theoretical understanding in your field. How do they support, challenge, or extend existing theories?

4.3. Contributions

4.2.2 Practical Implications

Discuss the practical applications of your research. How can your findings be used in real-world contexts?

4.3 Contributions

Clearly state the contributions of your research:

- Contribution to knowledge in your field
- Methodological contributions
- Practical contributions

4.4 Limitations

Acknowledge the limitations of your study. Be honest about what your research could not address and how these limitations might affect the interpretation of your results.

4.5 Future Research Directions

Based on your findings and limitations, suggest directions for future research:

- Areas that need further investigation
- Questions that emerged from your research
- Potential improvements to methodology
- Extensions of your work

4.6 Final Remarks

Conclude with final thoughts about your research and its significance. End on a strong note that emphasizes the value of your work.

Bibliography

Author, A. A., & Author, B. B. (Year). Title of the article. *Journal Name*, Volume(Issue), page-page.

Author, C. C. (Year). *Title of the book*. Publisher.

Author, D. D., Author, E. E., & Author, F. F. (Year). Title of the conference paper. In *Conference Proceedings Name* (pp. page-page). Publisher.

Author, G. G. (Year). Title of the thesis. [Doctoral dissertation or Master's thesis, Institution Name]. Database or Archive Name.

Organization Name. (Year). Title of the report. <<https://www.example.com/report>>(Accessed Month Year).

Author, H. H. (Year). Title of the web page. Website Name, <<https://www.example.com>>(Accessed Month Year).

Note: This is a manual bibliography format. For automatic bibliography management, consider using BibTeX with a .bib file. To use BibTeX:

1. Put your entries in `References.bib` (a starter file is included)
2. Set `\useBibTeX` to `true` in `config.tex`
3. Recompile with `compile.sh` or `compile.bat`

Nothing else changes: this page is replaced by the generated bibliography, and the `\cite` examples in Chapter 1 start working.

Example BibTeX entry:

```
@article{author2023,
```

```
author = {Author, A. A. and Author, B. B.},  
title = {Title of the Article},  
journal = {Journal Name},  
year = {2023},  
volume = {10},  
number = {2},  
pages = {123-145}  
}
```

Then cite in text using: `\cite{author2023}`

Appendices

Appendix A

Additional Data and Tables

This appendix contains additional data, tables, and figures that support the main text but are too detailed or lengthy to include in the main chapters.

A.1 Supplementary Tables

Table A.1: Supplementary data table

Item	Value 1	Value 2	Value 3
Data 1	10.5	12.3	14.7
Data 2	8.9	9.2	10.1
Data 3	15.2	16.8	17.3

A.2 Supplementary Figures

Include additional figures that provide supporting evidence or detailed visualizations.

Figure A.1: Supplementary figure showing additional results

A.3 Raw Data

If appropriate, include raw data or detailed measurements that were used in your analysis.

Appendix B

Code and Algorithms

This appendix contains code listings, algorithms, and technical implementation details.

B.1 Algorithm Pseudocode

You can include algorithm pseudocode here:

Algorithm: Example Algorithm

Input: data array, threshold value

Output: processed results

1. Initialize result array
2. For each element in data:
 - a. If element > threshold:
 - Add element to result
 - b. Else:
 - Skip element
3. Return result

B.2 Code Listings

Include relevant code snippets that support your research:

B.2.1 Python Example

Listing B.1: Example Python code

```
import numpy as np
import matplotlib.pyplot as plt

def process_data(data, threshold):
    """
    Process data based on threshold
    """
    result = []
    for value in data:
        if value > threshold:
            result.append(value)
    return np.array(result)

# Example usage
data = np.random.randn(100)
processed = process_data(data, 0.5)
```

B.2.2 R Example

Listing B.2: Example R code

```
# Load required libraries
library(ggplot2)

# Process data
process_data <- function(data, threshold) {
  result <- data[data > threshold]
  return(result)
```

```
}
```

```
# Example usage
```

```
data <- rnorm(100)
```

```
processed <- process_data(data, 0.5)
```

B.3 Configuration Files

Include any configuration files or parameter settings used in your research.